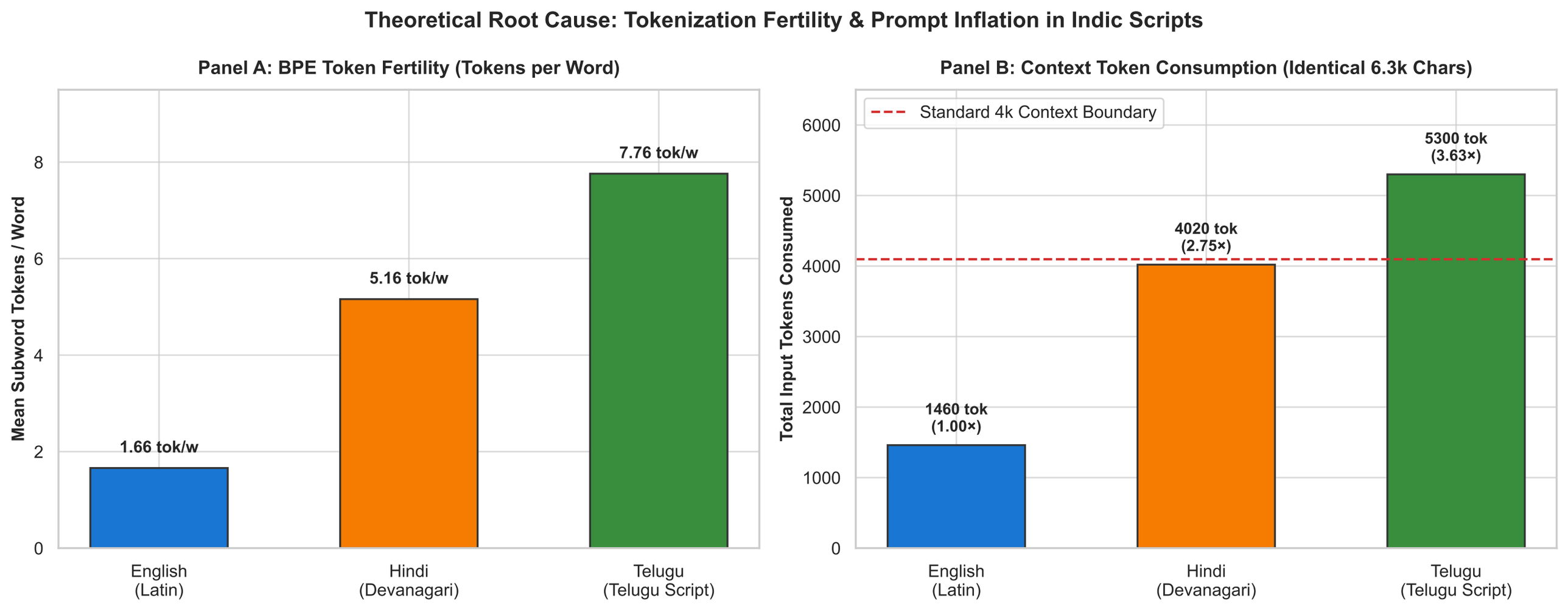
Theoretical Root Cause: Tokenization Fertility & Prompt Inflation in Indic Scripts
Panel A: BPE Token Fertility (Tokens per Word)
Mean Subword Tokens / Word
0
2
4
6
8
1.66 tok/w
5.16 tok/w
7.76 tok/w
English (Latin)
Hindi (Devanagari)
Telugu (Telugu Script)
Panel B: Context Token Consumption (Identical 6.3k Chars)
Total Input Tokens Consumed
0
1000
2000
3000
4000
5000
6000
Standard 4k Context Boundary
1460 tok (1.00×)
4020 tok (2.75×)
5300 tok (3.63×)
English (Latin)
Hindi (Devanagari)
Telugu (Telugu Script)